How Far Off the Recorded Path Can a Gaussian-Splat Driving Simulator Be Trusted? Mid-Term Report

CS 566 Project Mid-Term Report · Kyle Sha · Allen Xia · 29 Oct 2026

1. Summary of Progress

The core study described in the proposal is complete, and all extensions and stretch goals have at least a first result. We reconstruct Gaussian splats [13] of eight held-out Para-Lane [8] roads from one lane each, render them at the recorded poses of the neighboring lanes, and compare the renders with the real images captured there. The comparison covers the raw render, the released Difix [1], NVIDIA Fixer [4], our re-implementation of the Difix training procedure, the released Difix fine-tuned on our pairs (Difix-FT, added as a reference point), a small U-Net trained from scratch, the progressive 3D update of Difix3D+ [1], and a road-surface prior after ViSE [15]. Each method is scored for lane agreement with the real image (CLRNet [14] lane F1), perceptual distance (LPIPS), invented and erased lane paint, and spurious vehicles (YOLO11). All intervals are 95% bootstrap intervals over lane-pair sequences.

The status of each item is:

Tier	Item	Status
Core	Splats, raw render, released Difix, NVIDIA Fixer, evaluation framework	Complete (8 roads)
Core	Difix training re-implementation, validated against the official code and weights	Complete
Extensions	Small U-Net, road-surface prior, progressive update, more roads	Complete
Extensions	Heading axis	Complete on 2 roads
Stretch	Camera height (road-plane warp and MV2 car/scooter pairs)	Complete, with a caveat (Section 3)
Stretch	Cross-dataset (EUVS/MARS [7], MV2 [9]); FaithFusion [2]; closed loop in HUGSIM [3]	Complete
Added	Our own splat renderer, validated against gsplat	Complete

2. Current Results

Trust falls quickly off the recorded path. On the recorded path every method agrees with the real image (lane F1 about 0.97). For the raw render, lane F1 falls to 0.18 [0.13, 0.24] one lane over (median 3.0 m) and to 0.01 two lanes over (median 5.8 m) (Figure 1). A drop of similar size appears in KITScenes [16], where traffic-sign recall in a feed-forward reconstruction falls by about 80% at ±3 m, and RGE-GS [17] reports image-similarity scores for reconstruction methods at the same Para-Lane shifts.

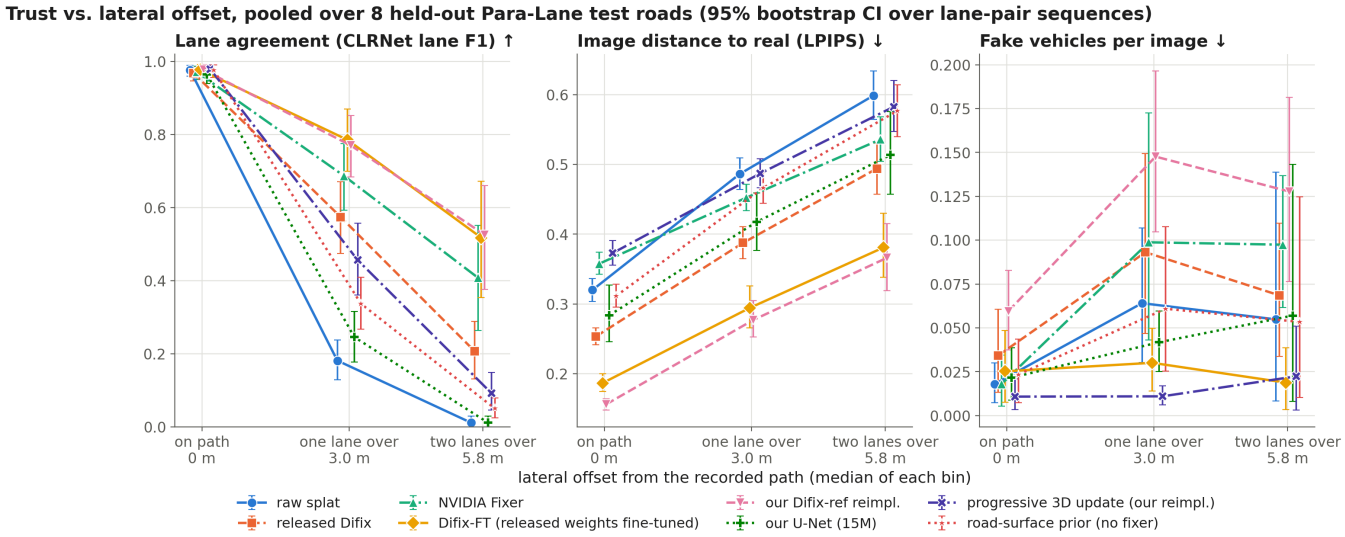


Figure 1. Lane agreement, perceptual distance and spurious vehicles against lateral offset, pooled over eight held-out roads.

Fixers recover part of the lane geometry, at a cost. Table 1 lists the main methods. The fixers trained on neighboring-lane pairs restore the most lane geometry. Our re-implementation, however, adds spurious vehicles on all eight roads (+0.089 per image [+0.062, +0.117] relative to the raw render), whereas Difix-FT does not. The progressive 3D update distills repaired views into the splat and improves lane F1 by +0.33 [+0.19, +0.49] across roads at about 3 m. Because the scene itself is refined, it produces the fewest spurious vehicles, but its gain at about 6 m is small. The road-surface prior involves no image generation and gives a smaller, consistent gain. The small U-Net stays close to the raw render.

Method	Lane F1, ~3 m	Lane F1, ~6 m	LPIPS, ~3 m	Spurious vehicles / image, ~3 m
Raw render	0.18 [0.13, 0.24]	0.01	0.49	0.064
Released Difix	0.57 [0.47, 0.67]	0.21	0.39	0.093
NVIDIA Fixer	0.69 [0.59, 0.78]	0.41	0.45	0.099
Difix-FT (released weights, fine-tuned on our pairs)	0.79 [0.70, 0.87]	0.52	0.29	0.030
Our Difix re-implementation	0.77 [0.68, 0.85]	0.53	0.28	0.148
Small U-Net (from scratch)	0.25 [0.18, 0.32]	0.01	0.42	0.042
Progressive 3D update	0.46 [0.36, 0.56]	0.09	0.49	0.011
Road-surface prior	0.34 [0.27, 0.41]	0.05	0.47	0.061

Table 1. Pooled results over eight roads (2,825 images at ~3 m, 1,386 at ~6 m).

Results vary strongly by road. The ranking of methods is stable, but the level is not: on a curved road the best fixer reaches lane F1 0.95 one lane over, while under an elevated highway with hard shadows no method exceeds 0.39 (Figure 2). Two lanes over on that road, every fixer produces a plausible road that does not match the real one.

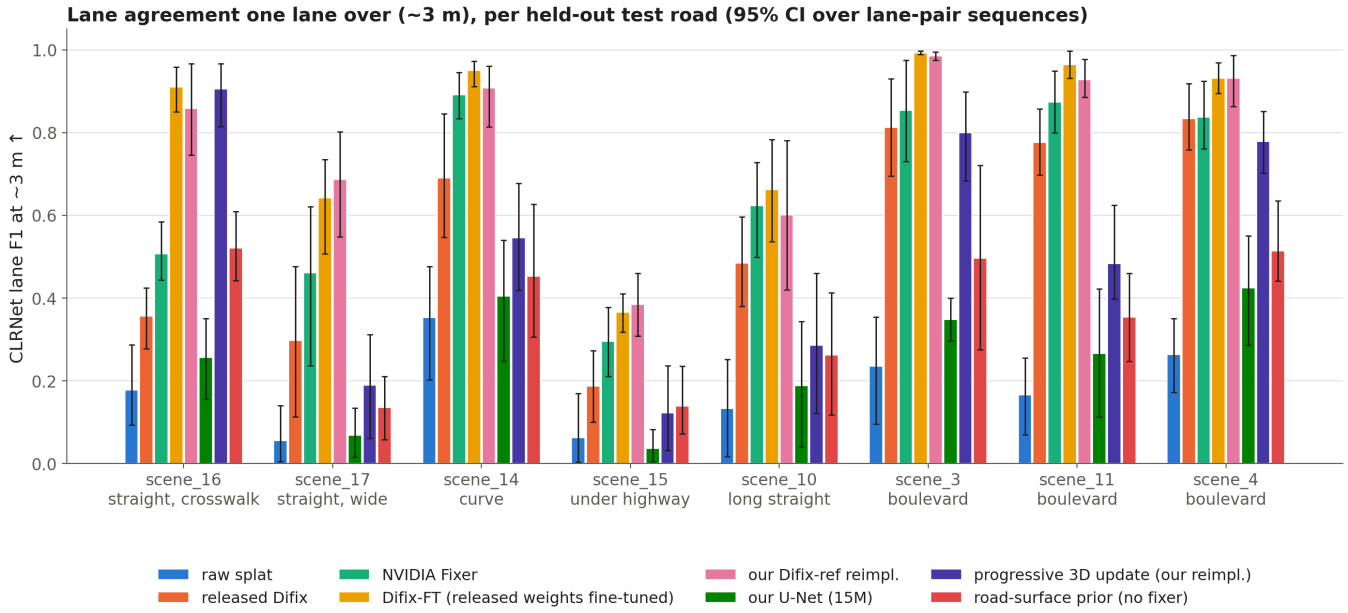


Figure 2. Lane agreement one lane over, per held-out road.

Invented content is visible and measurable. Figure 3 shows a typical case: our re-implementation paints a wide white marking that is absent from the real road (paint precision 0.13, compared with 0.95 for the raw render), while its lane F1 remains high.



Figure 3. Invented (orange) and erased (blue) lane paint, one lane over (4.1 m).

Re-implementation of the Difix training procedure. Starting from SD-Turbo with identical training pairs and 1,801 steps, our trainer outperforms the official trainer on DL3DV (LPIPS 0.218 versus 0.237, FID 26.1 versus 36.3; paired ΔLPIPS −0.019 [−0.020, −0.017]). With a reference view it matches NVIDIA's released Difix-ref on LPIPS (0.167 versus 0.163) and has a higher FID (21.2 versus 18.9). Only the DL3DV sample release was accessible, so this comparison uses three held-out scenes rather than the paper's benchmark.

Other axes and datasets. Turning the camera by up to ±15° changes lane F1 very little, whereas raising or lowering it by 1 m reduces road-region PSNR by 5–7 dB for the raw render. On the EUVS/MARS routes and the MV2 car/scooter pairs, the fixers trained on Para-Lane do not transfer: the generic released Difix-ref gives the largest LPIPS improvement, and on MV2 no method exceeds lane F1 0.09. In HUGSIM closed loop (released LTF policy, 25 scenarios, 3 seeds), no fixer improves the driving score (no fixer 0.408; released Difix −0.015 [−0.064, +0.034]; Difix-FT −0.005 [−0.054, +0.048]; NVIDIA Fixer −0.051 [−0.101, −0.004]), and the variation between policy seeds exceeds every fixer effect. Without a fixer, the policy spends 44% of its steps 1–4.5 m from the recorded trajectory and 29% more than 4.5 m from it, so the fixers are exercised at the offsets studied above. FaithFusion's per-pixel information-gain map partly predicts invented lane paint one lane over (AUROC 0.67–0.76), but using it to weight the progressive update made no measurable difference.

Our own renderer. We replaced gsplat's projection with our own PyTorch implementation (camera transform, covariance, EWA projection, spherical-harmonic color, depth), keeping only gsplat's tile-sorting and blending kernels. Renders agree with gsplat to 90.8 dB PSNR on average, gradients agree with cosine similarity of at least 0.99996, and a full scene refit scores within the variation between gsplat runs. Training is 1.44 times slower.

3. Difficulties Encountered

- *Official Difix training code.* The released trainer needed six fixes to run (undefined names, a missing attribute, an ignored step limit) and exceeded 24 GB of GPU memory with the reference view, so the matched comparison with our trainer uses the single-view setting.
- *Data access.* DL3DV granted only its sample release; EUVS's rotation-only setting requires nuPlan data under a separate license and was not run.
- *Ground truth across lanes.* The three Para-Lane lanes were driven at different times, so traffic differs between the render and the real image. Moving objects are masked in all metrics.
- *Camera height.* MV2's poses are normalized, so metric scale rests on an assumed car camera height of 1.40 m, and its two vehicles are also about 7.6 m apart laterally. Height is therefore evaluated mainly with a road-plane warp of the real image.

- *Lane-paint masks.* The color-and-contrast paint detector also responds to hard shadows and bright curbs on three roads, which makes paint precision and recall noisy there; lane F1 is unaffected.
- *Third-party software.* FaithFusion's full pipeline depends on an unreleased restorer, so we use its information-gain map within our progressive update. NVIDIA Fixer requires NVIDIA Transformer Engine, which we could not build, so we run it through a small compatibility layer. Other HUGSIM driving policies depend on outdated software, so closed-loop results use LTF only.
- *Small fixer.* The U-Net trained from scratch mostly blurs the render and does not transfer to other roads.
- *Renderer.* Our first renderer implementation was 12 times slower than gsplat; restructuring the small matrix products reduced this to 1.7 times per training step.

4. Changes to the Proposal

The core, extensions and stretch goals remain as proposed. We added two items: Difix-FT, the released Difix fine-tuned on our pairs, as a reference point between the released and re-implemented fixers; and our own splat renderer, validated against gsplat, so that the projection step of the 3D reconstruction is implemented by us. Camera height relies on the road-plane warp more than planned, and the heading axis covers two roads. The remaining work is to consolidate the analysis per road, prepare the final figures and videos, and complete the presentation and project webpage.

References

[1] Difix3D+, CVPR 2025 · arxiv.org/abs/2503.01774
[2] FaithFusion, CVPR 2026 · arxiv.org/abs/2511.21113
[3] HUGSIM, TPAMI 2026 · arxiv.org/abs/2412.01718
[4] NVIDIA Fixer, 2025 · github.com/nv-tlabs/Fixer
[7] EUVS benchmark, ICCV 2025 · arxiv.org/abs/2412.05256
[8] Para-Lane dataset, 3DV 2025 · arxiv.org/abs/2502.15635
[9] MV2 dataset, ECCV 2026 · arxiv.org/abs/2608.12442
[13] 3D Gaussian Splatting, SIGGRAPH 2023 · arxiv.org/abs/2308.04079
[14] CLRNet, CVPR 2022 · arxiv.org/abs/2203.10350
[15] ViSE, ICCV 2025 · arxiv.org/abs/2510.18341
[16] KITScenes, 2026 · arxiv.org/abs/2606.02956
[17] RGE-GS, 2025 · arxiv.org/abs/2506.22800

Reference numbers [1]–[15] follow the proposal; unused numbers refer to works cited only there.